HCIA-AI V4.0 Certification Practice Exam • Chapter 6

Module 6: Cutting-Edge AI Applications (100 Questions & Answer Key)

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Question 1

TRUE OR FALSE • 16 Points

Huawei's smart voice assistant Celia (Xiaoyi) integrates Automatic Speech Recognition (ASR), Natural Language Understanding (NLU), and Text-to-Speech (TTS) technologies.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

In autonomous driving systems, LiDAR sensors rely completely on capturing passive ambient visible sunlight and do not emit any laser light.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

Collaborative Filtering in recommendation systems relies only on the textual metadata keywords of items and completely ignores historical user behavior logs.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

Huawei's Qiankun Intelligent Driving solution provides full-stack autonomous driving capabilities covering urban scenarios, highways, and automated valet parking.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

The Cold Start problem in recommendation systems refers to server processors running at low temperatures during winter.

A. True
B. False

Page Answers: **Q1**: True • **Q2**: False • **Q3**: False • **Q4**: True • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

In intelligent vehicles, HarmonyOS Cockpit enables seamless cross-device interconnection between smartphones, smart watches, and vehicle infotainment displays.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

Driver Monitoring Systems (DMS) are designed to monitor tire pressure and engine oil levels, having nothing to do with driver fatigue or attention.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

Embodied Artificial Intelligence focuses on developing intelligent agents that perceive physical environments through multimodal sensors and physically interact using robotic actuators.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

In AI4Science, Huawei's Pangu-Weather model relies only on manual handwritten differential equations without using any deep neural networks.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

In smart homes, AI-enabled cameras with acoustic detection can recognize abnormal events such as glass breaking or infant crying to trigger security alarms.

A. True
B. False

Page Answers: **Q6**: True • **Q7**: False • **Q8**: True • **Q9**: False • **Q10**: True

Question 11

TRUE OR FALSE • 16 Points

Simultaneous Localization and Mapping (SLAM) is a technology used only for chemical synthesis and cannot be applied to robot navigation.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

Item-based collaborative filtering recommends items to a user based purely on the user's demographic age and gender without analyzing item-item interaction history.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

End-to-End (E2E) autonomous driving directly maps raw sensor inputs (e.g., multi-camera video streams) into vehicle trajectory and control outputs using a unified deep neural network.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

In AI + Molecular Dynamics, neural network potentials enable atomic simulations over long biological timescales with quantum-mechanical level accuracy.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

Smart home climate control systems only support hard-coded manual thermostat switches and cannot dynamically adapt to occupant presence or outdoor weather patterns.

A. True
B. False

Page Answers: **Q11**: False • **Q12**: False • **Q13**: True • **Q14**: True • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

Keyword Spotting (KWS) in voice assistants requires continuous cloud-based training of 100-billion parameter models for every millisecond of background audio.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

Sensor fusion in intelligent driving combines data from multiple complementary sensors (cameras, millimeter-wave radars, LiDAR) to overcome the environmental limitations of any individual sensor.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

User-based collaborative filtering assumes that users with similar historical ratings will have completely opposite preferences for future items.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

AI-assisted materials science can screen millions of virtual crystal compositions to identify candidate compounds for novel solid-state batteries.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

Autonomous mobile robots (AMR) in smart warehouses rely only on physical railway tracks and cannot perform dynamic obstacle avoidance.

A. True
B. False

Page Answers: **Q16**: False • **Q17**: True • **Q18**: False • **Q19**: True • **Q20**: False

Question 21

TRUE OR FALSE • 16 Points

In natural language dialogue management, Dialogue State Tracking (DST) maintains the current state and contextual slot values across multiple turns of user conversation.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Millimeter-wave (mmWave) radars are severely degraded by light rain or fog and cannot detect vehicle velocities.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

In AI + Quantum Chemistry, machine learning models can predict molecular electronic properties and energy states significantly faster than traditional Density Functional Theory (DFT).

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Content-based recommendation algorithms recommend items only based on popularity rankings across the general population without considering item attributes.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

Modern intelligent robots use Large Multimodal Models (LMMs) to ground abstract high-level linguistic commands into sequence-level physical robotic action primitives.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: True • **Q24**: False • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an advanced AI application domain?

- **A.** Autonomous vehicular path trajectory planning
- **B.** Medical radiological diagnostic segmentation
- **C.** Traditional relational database B-tree indexing
- **D.** Intelligent power grid dispatch optimization

Question 27

SINGLE-ANSWER QUESTION • 16 Points

In Huawei's smart voice assistant architecture, which component is responsible for understanding the user's underlying intent and extracting key entity slots from transcribed text?

- **A.** Automatic Speech Recognition System
- **B.** Natural Language Understanding Engine
- **C.** Text-to-Speech Acoustic Synthesizer
- **D.** Dialog State Management Controller

Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which sensor in an autonomous driving vehicle is most effective at directly capturing high-resolution color, texture, and text details of traffic signs and signal lights?

- **A.** Millimeter-wave Radar
- **B.** Optical HD Camera
- **C.** Ultrasonic Sensor
- **D.** Wheel Speed Encoder

Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which sensor emits radio waves and uses the Doppler effect to measure the precise relative radial velocity of moving objects, even in adverse weather like fog or heavy rain?

- **A.** Monocular RGB Optical Camera Sensor
- **B.** Ultrasonic Close-Range Proximity Sensor
- **C.** Millimeter-wave Radar Unit
- **D.** Micro-Electro-Mechanical Gyroscope

Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which recommendation approach recommends products to User A by finding other users whose past rating profiles are mathematically closest to User A?

- **A.** Random Permutation Generator
- **B.** User-Based Collaborative Filtering
- **C.** Item-Based Collaborative Filtering
- **D.** Content-Based Metadata Matching

Page Answers: **Q26**: C • **Q27**: B • **Q28**: B • **Q29**: C • **Q30**: B

Question 31

SINGLE-ANSWER QUESTION • 16 Points

When an e-commerce platform launches a brand-new product with zero customer reviews, clicks, or purchase records, which challenge does it face?

- **A.** Catastrophic Forgetting
- **B.** Pipeline Parallelism Stall
- **C.** Vanishing Gradient Problem
- **D.** Item Cold Start Problem

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Huawei's flagship commercial smart voice assistant available across its HarmonyOS device ecosystem is known as:

- **A.** Celia
- **B.** Siri
- **C.** Alexa
- **D.** Bixby

Question 33

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the Planning & Decision-Making system in autonomous driving?

- **A.** Generating collision-free trajectories and behavioral actions based on fused perception data
- **B.** Extracting 3D bounding boxes and semantic lane markings from raw camera and LiDAR sensors
- **C.** Estimating centimeter-level vehicle pose relative to high-definition prior spatial maps
- **D.** Converting target trajectory commands into low-level steering angle and braking pressure signals

Question 34

SINGLE-ANSWER QUESTION • 16 Points

What is the core function of SLAM (Simultaneous Localization and Mapping) in autonomous robotics?

- **A.** Executing global route navigation across pre-computed static highway road networks
- **B.** Estimating robot spatial pose while building a consistent map of an unknown surrounding environment
- **C.** Regulating closed-loop PID servo motor torque across robotic manipulator joints
- **D.** Classifying static road traffic signs using pre-trained 2D convolutional networks

Question 35

SINGLE-ANSWER QUESTION • 16 Points

In smart home ecosystems, what is the primary benefit of Edge AI processing on smart hubs?

- **A.** Ensuring low-latency inference and local user privacy protection without constant cloud offloading
- **B.** Offloading all raw audiovisual streams to remote public cloud data centers
- **C.** Requiring continuous gigabit internet bandwidth for routine offline tasks
- **D.** Replacing local neural inference chips with remote cloud virtual machines

Page Answers: **Q31**: D • **Q32**: A • **Q33**: A • **Q34**: B • **Q35**: A

Question 36

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei foundation model has demonstrated groundbreaking achievements in high-speed, accurate global medium-range numerical weather prediction?

- **A.** Pangu-Weather
- **B.** Pangu-Mining
- **C.** Pangu-Code
- **D.** Pangu-Finance

Question 37

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of Text-to-Speech (TTS) in an intelligent voice assistant?

- **A.** Segmenting audio files into 1-second chunks
- **B.** Encrypting Wi-Fi passwords
- **C.** Converting structured response text into natural, fluent human-like spoken audio
- **D.** Listening to spoken microphone input and transcribing it into text

Question 38

SINGLE-ANSWER QUESTION • 16 Points

Which technology enables modern voice assistants to listen and recognize user voice commands across noisy living rooms?

- **A.** Far-field microphone arrays combined with acoustic beamforming, dereverberation, and noise suppression
- **B.** Optical infrared cameras detecting micro-vibrations across human vocal cords through window glass
- **C.** Rule-based symbolic syntax parsing using context-free grammar trees
- **D.** Mechanical diaphragm springs that physically amplify sound waves using resonance chamber cones

Question 39

SINGLE-ANSWER QUESTION • 16 Points

What level of autonomous driving (according to SAE J3016) represents 'High Driving Automation' where the vehicle handles all driving tasks under specific operational design domains (ODD) without expecting driver intervention?

- **A.** Level 4
- **B.** Level 1
- **C.** Level 0
- **D.** Level 2

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the 'Smart Cockpit' in intelligent vehicles?

- **A.** The physical aerodynamic rear spoiler adjusting vehicle downforce at high highway cruising speeds
- **B.** Heavy-duty cast-iron engine mounting brackets absorbing rotational powertrain vibrations
- **C.** In-cabin multimodal infotainment displays, HUD, voice control, and driver monitoring systems
- **D.** Hydraulic brake caliper fluid pressure reservoirs regulating emergency stopping distances

Page Answers: **Q36**: A • **Q37**: C • **Q38**: A • **Q39**: A • **Q40**: C

Question 41

SINGLE-ANSWER QUESTION • 16 Points

How does AI accelerate drug discovery in pharmaceutical science?

- **A.** Automating plastic mold injection machines manufacturing pill bottles in fulfillment warehouses
- **B.** Replacing living human clinical trial volunteers with basic mathematical spreadsheet formulas
- **C.** Scheduling cargo airplane delivery flight paths for commercial wholesale pharmaceutical shipments
- **D.** Predicting protein 3D structures, molecular docking affinity, and optimizing candidate chemical molecules

Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which technology allows an autonomous vehicle to maintain accurate sub-meter positioning in urban canyons?

- **A.** Dead-reckoning wheel odometry without external spatial map correction
- **B.** Single-frequency civil GPS positioning without inertial sensor fusion
- **C.** Monocular optical flow estimation without metric depth calibration
- **D.** High-Definition Map matching fused with multi-sensor GNSS-RTK and LiDAR visual odometry

Question 43

SINGLE-ANSWER QUESTION • 16 Points

What does 'Embodied AI' mainly require that distinguishes it from purely virtual AI software?

- **A.** Training only on unlabeled text datasets
- **B.** Replacing self-attention layers with static 1D max-pooling operations
- **C.** Registered corporate developer accounts on public social media web platforms and message boards
- **D.** A physical agent interacting with the real physical world through sensors and physical actuators

Question 44

SINGLE-ANSWER QUESTION • 16 Points

In a smart home, what type of AI sensor can detect human occupancy and breathing without optical cameras?

- **A.** Millimeter-wave Radar Micro-Doppler Motion Sensors
- **B.** Atmospheric barometric altitude pressure measurement transducers
- **C.** Passive photoresistor cadmium-sulfide ambient illuminance sensors
- **D.** Passive monocular RGB cameras without active illumination

Question 45

SINGLE-ANSWER QUESTION • 16 Points

Which deep learning recommendation model introduced by Google combines a linear generalized model (for memorization) with a deep feedforward network (for generalization)?

- **A.** Pure K-Means
- **B.** LeNet-5
- **C.** Wide & Deep Model
- **D.** Single Decision Tree

Page Answers: **Q41**: D • **Q42**: D • **Q43**: D • **Q44**: A • **Q45**: C

Question 46

SINGLE-ANSWER QUESTION • 16 Points

In voice assistants, what is the role of Keyword Spotting (KWS)?

- **A.** Continuous low-power acoustic monitoring to detect pre-configured wake words
- **B.** Translating multi-hour foreign language cinema dialogue scripts into localized subtitle files
- **C.** Synthesizing full audiobooks from raw plain-text electronic manuscripts using neural vocoders
- **D.** Measuring indoor air quality and room humidity levels using solid-state chemical sensors

Question 47

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of Ultrasonic Sensors in modern production cars?

- **A.** Short-range obstacle distance measurement during low-speed automated parking maneuvers
- **B.** Offline batch sentiment classification of customer review datasets
- **C.** Recognizing highway speed limit traffic signs and overhead optical electronic toll gantries
- **D.** Tracking high-speed oncoming vehicles at three hundred meters in dense highway fog

Question 48

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a classic technique used to solve the User Cold-Start problem in recommendation systems?

- **A.** Blanking the consumer screen completely until the user performs ten physical purchase transactions
- **B.** Requiring new account registrants to solve five complex multi-variable calculus equations
- **C.** Presenting an interactive onboarding interest selection questionnaire to capture initial user preferences
- **D.** Automatically terminating inactive account sessions after thirty seconds of scrolling dormancy

Question 49

SINGLE-ANSWER QUESTION • 16 Points

In robotics, what is the computational problem of calculating joint motor angles required to position an end-effector gripper at a target coordinate?

- **A.** Gradient Descent
- **B.** Inverse Kinematics
- **C.** Data Augmentation
- **D.** Batch Normalization

Question 50

SINGLE-ANSWER QUESTION • 16 Points

How does AI + Quantum Mechanics assist in materials science?

- **A.** Exact analytical closed-form solution of multi-electron Schrödinger equations without approximation
- **B.** Substituting physical experimental verification with static database lookup tables
- **C.** Applying macroscopic Newtonian rigid-body mechanics to quantum wavepacket states
- **D.** Predicting electronic ground-state energies and molecular wavefunctions via neural approximations

Page Answers: **Q46**: A • **Q47**: A • **Q48**: C • **Q49**: B • **Q50**: D

Question 51

SINGLE-ANSWER QUESTION • 16 Points

Which component in dialogue systems decides what action the system should take next?

- **A.** Dialogue Policy
- **B.** Speech Recognition
- **C.** Intent Classifier
- **D.** Speech Synthesis

Question 52

SINGLE-ANSWER QUESTION • 16 Points

What is 'V2X' (Vehicle-to-Everything) communication in intelligent transportation?

- **A.** Eliminating electronic brake sensor communication buses
- **B.** Wireless data exchange between vehicles, roadside units, pedestrians, and network infrastructure
- **C.** Executing offline k-means clustering on static vehicle registration tables
- **D.** Retrofitting internal combustion engines with commercial compressed liquid natural gas cylinders

Question 53

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes 'Matrix Factorization' in collaborative filtering recommendation systems?

- **A.** Inverting matrices to solve linear regression equations
- **B.** Joining tables across relational databases using primary keys
- **C.** Applying convolutional filters to rotate image feature maps
- **D.** Decomposing the user-item interaction matrix into lower-dimensional user and item feature vectors

Question 54

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei intelligent vehicle solution provides high-precision navigation and autonomous driving capabilities?

- **A.** HiSilicon Multimedia Video Decoder Service
- **B.** MindSpore Model Zoo Hub Repository
- **C.** Huawei Qiankun ADS
- **D.** ModelArts Lite Edge Container Daemon

Question 55

SINGLE-ANSWER QUESTION • 16 Points

In intelligent robotics, what is the role of 'Tactile Sensors' integrated into robotic grippers?

- **A.** Measuring long-range obstacle distances using time-of-flight laser pulses
- **B.** Estimating rotational angular velocity and linear acceleration of the robot chassis
- **C.** Capturing wide-angle stereoscopic RGB depth maps of the surrounding workspace
- **D.** Measuring contact forces, shear slip, and pressure distributions during delicate object manipulation

Page Answers: **Q51**: A • **Q52**: B • **Q53**: D • **Q54**: C • **Q55**: D

Question 56

SINGLE-ANSWER QUESTION • 16 Points

What is the primary objective of Occupancy Grid Mapping in autonomous navigation?

- **A.** Organizing unstructured text log records chronologically inside cloud object storage buckets
- **B.** Offline batch backpropagation across historical server log archives
- **C.** Assigning numbered reserved seating tickets across municipal public transportation buses
- **D.** Discretizing surrounding spatial terrain into grid cells representing the probability of obstacles

Question 57

SINGLE-ANSWER QUESTION • 16 Points

In smart buildings, what does AI-driven HVAC optimization achieve?

- **A.** Replacing physical metal ventilation ductwork with wireless digital microwave communication links
- **B.** Dynamically adjusting thermal heating and cooling based on real-time occupancy and weather forecasts
- **C.** Manually rotating mechanical window blind louvers based on printed solar equinox almanac charts
- **D.** Overclocking building electricity consumption by running redundant industrial refrigeration units

Question 58

SINGLE-ANSWER QUESTION • 16 Points

What enables modern humanoid robots to understand complex unstructured natural language commands and act?

- **A.** Hard-coded sequential conditional if-else statements executed by programmable logic controllers
- **B.** Vision-Language-Action foundation models translating multimodal perception into robotic actions
- **C.** Purely reactive open-loop motor commands without visual or semantic scene grounding
- **D.** Classical PID single-variable feedback controllers operating without language interpretation

Question 59

SINGLE-ANSWER QUESTION • 16 Points

In recommendation systems, what does 'Popularity Bias' refer to?

- **A.** Ranking all catalog items strictly by reverse chronological upload timestamp
- **B.** Displaying discount promotional items only to verified enterprise corporate account holders
- **C.** Suppressing item recommendations completely for users who exhibit high daily browsing activity
- **D.** The algorithmic tendency to over-recommend viral head items while ignoring the diverse long tail

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Huawei Pangu-Weather AI model are correct? (Multiple Choice)

- **A.** It employs a 3D Earth-Specific Transformer architecture to process 3D atmospheric variables
- **B.** It generates global weather forecasts at hour-level precision in just seconds on Ascend NPUs
- **C.** Its forecasting precision surpasses traditional Numerical Weather Prediction systems across key metrics
- **D.** It relies completely on 2D convolutional edge-detection kernels without temporal modeling

Page Answers: **Q56**: D • **Q57**: B • **Q58**: B • **Q59**: D • **Q60**: ABC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common application scenarios and tasks of Natural Language Generation (NLG)?

- **A.** Automated multi-document summarization and executive report drafting
- **B.** Syntactic part-of-speech tagging and dependency tree parsing
- **C.** Conversational dialogue agent response generation in customer support chatbots
- **D.** Lexical named entity boundary extraction and classification

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

In Autonomous Driving perception systems, which spatial parameters are estimated for detected 3D bounding boxes?

- **A.** 3D spatial center coordinates relative to vehicle sensor coordinate origin
- **B.** Physical object dimensional extents
- **C.** Camera intrinsic focal length and optical lens radial distortion calibration matrix
- **D.** Object orientation yaw angle and estimated 3D velocity vector

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following domain applications naturally benefit from Graph Neural Networks (GNNs) due to their non-Euclidean relational structure?

- **A.** Uniform 2D pixel grid convolution on standard grayscale bitmap images
- **B.** Molecular chemical property prediction and drug-target interaction modeling
- **C.** Financial fraud ring detection and anti-money laundering in transaction graphs
- **D.** Social network community clustering, user influence modeling, and citation analysis

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In intelligent vehicles, which vehicular sub-systems are commonly actuated through Drive-by-Wire (DbW) electronic control signals?

- **A.** Steer-by-Wire
- **B.** Passive suspension
- **C.** Brake-by-Wire
- **D.** Throttle-by-Wire

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent recognized, high-impact clinical and medical applications of Artificial Intelligence in Healthcare?

- **A.** Radiological lesion detection, organ segmentation, and anomaly screening in CT and MRI scans
- **B.** Genomic variant pathogenicity prediction, mutation risk scoring, and personalized oncology analysis
- **C.** Automated digital pathology slide whole-image tumor tissue grading and classification
- **D.** AI-assisted molecular screening for de novo drug discovery and target-protein binding affinity prediction

Page Answers: **Q61**: AC • **Q62**: ABD • **Q63**: BCD • **Q64**: ACD • **Q65**: ABCD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following core functional modules constitute a traditional modular voice assistant system (such as Huawei Celia)?

- **A.** Automatic Speech Recognition
- **B.** Text-to-Speech
- **C.** Dialogue Management and Natural Language Understanding
- **D.** Unsupervised optical flow motion vector estimation

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary sensors used in autonomous driving perception systems?

- **A.** Light Detection and Ranging
- **B.** Millimeter-Wave Radar
- **C.** Ultrasonic Sensors
- **D.** Optical HD Cameras

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical AI-powered capabilities deployed in Smart Home environments?

- **A.** Intelligent voice-activated appliance control and conversational queries
- **B.** Smart climate control and energy optimization based on occupancy patterns
- **C.** Central bank interbank financial settlement transaction clearing
- **D.** Acoustic anomaly detection

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent classic main types of recommendation algorithms?

- **A.** Random Permutation Shuffling
- **B.** Collaborative Filtering
- **C.** Content-Based Filtering
- **D.** Reverse Lexical Alphabetical Sorting

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Sensor Fusion in autonomous driving are correct?

- **A.** Millimeter-wave radar directly measures target velocity and operates reliably in fog, rain, and snow
- **B.** Combining multiple sensors overcomes the physical limitations of individual sensors
- **C.** LiDAR provides direct, highly accurate 3D spatial coordinate measurements regardless of ambient lighting
- **D.** Cameras provide rich visual color, semantic text, and high angular resolution

Page Answers: **Q66**: ABC • **Q67**: ABCD • **Q68**: ABD • **Q69**: BC • **Q70**: ABCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following scientific fields are actively transformed by AI for Science (AI4Science)?

- **A.** Quantum mechanical electronic energy and molecular wavefunction approximations
- **B.** Novel materials discovery and crystalline structure simulation
- **C.** Global medium-range numerical weather and atmospheric climate prediction
- **D.** High-frequency relational database transaction concurrency locking

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are capabilities supported by Huawei's HarmonyOS Smart Cockpit?

- **A.** Driver fatigue and distraction monitoring
- **B.** Direct physical wireless override of municipal traffic intersection signal controllers
- **C.** Multi-screen sharing and phone-to-car seamless navigation continuation
- **D.** Intelligent voice dialogue and in-cabin multi-zone sound pickup

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following strategies can effectively solve the Cold Start problem in recommendation systems?

- **A.** Completely barring newly registered users from accessing platform content
- **B.** Using demographic onboarding preference questionnaires during user registration
- **C.** Using item metadata, attributes, and tags for content-based recommendations
- **D.** Recommending globally popular, highly-rated, or trending items to new users

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following tasks belong to the Perception layer rather than the Planning or Control layers in an autonomous vehicle?

- **A.** 3D Object Detection
- **B.** Lane Boundary Semantic Segmentation
- **C.** Collision-Free Trajectory Optimization
- **D.** Steering Actuator Torque Regulation

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Intelligent Robots and Embodied AI are correct?

- **A.** They rely on multimodal foundation models to interpret real-world commands
- **B.** They integrate perception, high-level reasoning, and physical manipulation
- **C.** They process static tabular datasets without real-world sensor-actuator interaction
- **D.** They require spatial reasoning and dynamics control to interact with physical objects

Page Answers: **Q71**: ABC • **Q72**: ACD • **Q73**: BCD • **Q74**: AB • **Q75**: ABD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized technical advantages of Huawei's Pangu-Weather model over traditional Numerical Weather Prediction (NWP)?

- **A.** Eliminates all physical radar and satellite observation sensors
- **B.** 10,000x faster inference speed on Ascend AI hardware
- **C.** Superior forecast accuracy on extreme weather events like tropical cyclone tracks
- **D.** Operates with zero historical meteorological observation data

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

In robotics, SLAM mainly solves which two tasks at the same time in an unknown environment?

- **A.** Localization
- **B.** Path planning
- **C.** Mapping
- **D.** Motion control

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common interaction modalities supported in modern intelligent cockpits?

- **A.** Touchscreen gesture control
- **B.** Vision-based contactless air gestures
- **C.** Natural voice interaction
- **D.** Driver gaze and eye-tracking interaction

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are benefits of deploying AI in Industrial Manufacturing and Smart Logistics?

- **A.** Autonomous mobile robots for automated material handling and palletizing
- **B.** Predictive equipment maintenance based on sensor vibration monitoring
- **C.** Guaranteeing zero inventory warehouse management
- **D.** Automated optical inspection for high-speed product defect detection

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are metrics commonly used to evaluate recommendation systems?

- **A.** Normalized Discounted Cumulative Gain
- **B.** Click-Through Rate and Conversion Rate
- **C.** Mean Absolute Error in rating prediction
- **D.** Precision@K and Recall@K

Page Answers: **Q76**: BC • **Q77**: AC • **Q78**: ABCD • **Q79**: ABD • **Q80**: ABCD

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key challenges in Autonomous Driving?

- **A.** Handling edge cases and unexpected road hazards
- **B.** Making vehicles invisible to human vision
- **C.** Operating reliably in extreme weather conditions
- **D.** Ensuring high safety redundancy across hardware, sensors, and computing compute units

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Collaborative Filtering algorithms are mainly divided into which classic memory-based approaches?

- **A.** Quantum Tunneling Filtering
- **B.** User-Based Collaborative Filtering
- **C.** Item-Based Collaborative Filtering
- **D.** Solid-State Cache Sector Filtering

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are safety features enabled by AI in Smart Home security?

- **A.** Visual perimeter intrusion detection with human-vs-pet classification
- **B.** Facial recognition on smart doorbells to identify family members vs strangers
- **C.** Acoustic detection of glass breakage or smoke detector alarms
- **D.** Elderly fall detection and automated emergency alert dispatch

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical autonomous driving scenarios supported by Huawei Qiankun ADS?

- **A.** Urban Navigation Cruise Assist
- **B.** Automated Valet Parking and Remote Parking
- **C.** Offline Tabular Data Warehouse ETL Indexing
- **D.** Highway Navigation Cruise Assist

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

In voice assistant systems, Natural Language Understanding (NLU) primarily performs which semantic extraction tasks?

- **A.** Intent classification
- **B.** Slot filling
- **C.** Speech synthesis
- **D.** Echo cancellation

Page Answers: **Q81**: ACD • **Q82**: BC • **Q83**: ABCD • **Q84**: ABD • **Q85**: AB

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements accurately characterize End-to-End Autonomous Driving compared to traditional modular pipelines?

- **A.** Directly maps raw sensor inputs to trajectory planning/control via unified neural networks
- **B.** Reduces compounding error accumulation across hand-engineered intermediate interfaces
- **C.** Eliminates the requirement for any sensor training data
- **D.** Uses large-scale multi-task driving demonstrations to learn complex driving behaviors

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent high-impact application scenarios of Artificial Intelligence in environmental protection and biodiversity conservation?

- **A.** Acoustic detection of illegal chainsaw logging and poaching vehicles in protected rainforests
- **B.** Satellite imagery analysis to monitor deforestation rates, illegal mining, and glacial retreat
- **C.** Autonomous underwater vehicle visual surveying of coral reef bleaching and marine biodiversity
- **D.** Camera-trap computer vision recognition to track and monitor endangered wildlife populations

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

In industrial and enterprise recommendation systems, which stages constitute the standard two-stage funnel architecture?

- **A.** Retrieval stage
- **B.** Ranking stage
- **C.** Data collection stage
- **D.** Feature labeling stage

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent core sensory-motor capabilities required for Embodied AI humanoid robots?

- **A.** Multimodal Environmental Perception
- **B.** Physical Motor Actuation and Coordinated Bipedal Locomotion
- **C.** Formal Axiomatic Mathematical Theorem Proving
- **D.** Whole-Body Dynamic Balance and Real-Time Trajectory Control

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major components of Huawei's All-Scenario AI Strategy?

- **A.** MindSpore full-scenario AI computing framework
- **B.** Ascend series AI processors providing scalable compute
- **C.** ModelArts platform enabling end-to-end AI lifecycle management
- **D.** Unified architecture across consumer devices, enterprise edge, and public/private clouds

Page Answers: **Q86**: ABD • **Q87**: ABCD • **Q88**: AB • **Q89**: ABD • **Q90**: ABCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

In intelligent voice interaction systems (AI voice assistants), which two speech processing technologies are mainly required?

- **A.** Automatic Speech Recognition
- **B.** Text-to-Speech synthesis
- **C.** Semantic Image Segmentation
- **D.** Machine Translation Alignment

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized functional sub-disciplines of Speech and Acoustic Signal Processing?

- **A.** Speech Recognition
- **B.** Speaker Identification and Voiceprint Verification
- **C.** Speech Enhancement and Acoustic Noise Suppression
- **D.** Spoken Language Understanding and Acoustic Wake-Word Spotting

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's official smart voice assistant across the HarmonyOS ecosystem is named ____.

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In autonomous driving, the optical sensor that uses pulsed lasers to measure distances and generate 3D point clouds is called ____.

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

In robotics and autonomous vehicles, SLAM stands for Simultaneous Localization and ____.

Answer: _____

Page Answers: **Q91**: AB • **Q92**: ABCD • **Q93**: Celia (or Xiaoyi) • **Q94**: LiDAR • **Q95**: Mapping

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

The challenge of providing accurate recommendations for new users or items with zero historical interaction is called the ____ Start problem.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's intelligent driving solution for high-level autonomous driving across highways, urban roads, and parking is called Huawei ____ (or ADS).

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's large meteorological model for high-precision global weather forecasting is named Pangu-____.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

The field of artificial intelligence that integrates physical bodies, sensors, and actuators to interact with physical environments is known as ____ AI.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In vehicle cockpits, DMS stands for Driver Monitoring System, which primarily detects driver ____ and distraction.

Answer: _____

Page Answers: **Q96**: Cold • **Q97**: Qiankun • **Q98**: Weather • **Q99**: Embodied • **Q100**: fatigue